You need to:

- **Calculate with squares, square roots, cubes and cube roots and other powers and roots of numbers.**

The **square** of a number n is $n \times n = n^2$, so the square of 5 is $5^2 = 25$

The **cube** of a number n is $n \times n \times n = n^3$, so the cube of 2 is $2^3 = 8$

$25 = 5^2$
5 is the **square root** of 25. The square root of n is represented by $\sqrt{n}$. So $\sqrt{25} = 5$

$8 = 2^3$
2 is the **cube root** of 8. The cube root of n is represented by $\sqrt[3]{n}$. So $\sqrt[3]{8} = 2$



on some calculators



on some calculators



on some calculators



on some calculators

Apply

Investigate how to enter powers and roots on your scientific calculator.

Recap

$2^4 = 16$ so $\sqrt[4]{16} = 2$

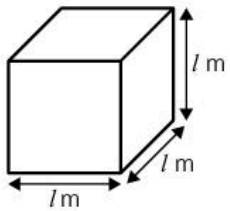
$3^5 = 243$ so $\sqrt[5]{243} = 3$

Key skills

You need to be able to work with other powers and roots, for example powers of 4 and fourth roots.

Worked example

A cube of side l metres has a volume of 30 cubic metres.



(a) Calculate the value of l.
(b) Calculate the area of one face of the cube. **[3 marks]**

(a) $l = \sqrt[3]{30} = 3.107...$
$= 3.11$ metres (to 3 s.f.)

(b) $(3.107...)^2 = 9.654...$
$= 9.65$ square metres (to 3 s.f.)

Exam tip

Use your calculator to find the cube root of 30 since $l^3 = 30$.

In part (b), use the *exact* answer from part (a) rather than the rounded answer. This ensures no rounding errors are carried forwards.

Exam tip

In a question like this, you will normally get one mark for evidence that you have entered the correct calculation and one mark for giving your answer to the correct level of accuracy (as specified in the question).

Worked example

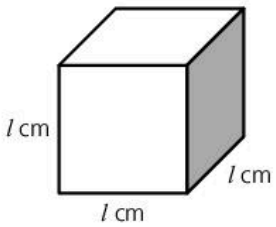
Use your calculator to work out $6^4 \div \sqrt[6]{1000}$. Round your answer to two decimal places. **[2 marks]**

$6^4 \div \sqrt[6]{1000} = 409.831...$

$= 409.83$ (to 2 d.p.)

Questions

1 Find l (to 3 s.f.) given that the volume of the cube is 50 cm^3.



2 Calculate the value of each expression (to 3 s.f. where necessary):

a 8^2 **b** 11^3 **c** 7.1^2

d $\sqrt{17}$ **e** $\sqrt[3]{-100}$ **f** $\sqrt[3]{6859}$

3 Find the side length of a square whose area is 49 cm^2.

4 Find the side length of a cube whose volume is 216 cm^3.

5 Write the following in ascending order (smallest first):

a 0.6^3 $\sqrt[3]{0.6}$ $\sqrt{0.6}$ 0.6 0.6^2

b 6^3 $\sqrt[3]{6}$ $\sqrt{6}$ 6 6^2

6 Calculate the value of each expression, rounding to 3 s.f. where appropriate.

a $3^2 \times \sqrt[4]{16}$

b $2^5 \div \sqrt[3]{8}$

c $5^4 \times \sqrt{7}$

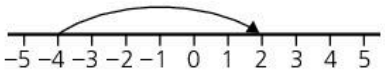
d $4^6 \div \sqrt[5]{20}$

You need to:

- **Use directed numbers in practical situations.**

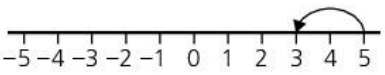
To add a positive number, or subtract a negative number, move to the right on a number line.

$(-4) + 6 = 2$



To subtract a positive number, or add a negative number, move to the left on a number line.

$5 + (-2) = 5 - 2 = 3$



Multiplying or dividing two negative numbers or two positive numbers gives a positive number.

Multiplying or dividing a negative number and a positive number gives a negative number.

$(-4) \times (-2) = 8$ $\quad 3 \times (-5) = -15$

$(-10) \div (-2) = 5$ $\quad (-6) \div 3 = -2$

Watch out!

You can remember the rule 'two negatives make a positive' when multiplying or dividing.

When adding or subtracting, the same rule does not work so use the number line method.

Worked example

In June, the average temperature in Moscow was 14°C.

In January, the average temperature was 25°C lower than in June.

What was the average temperature in Moscow in January?

[2 marks]

$14 - 25 = -11$°C

Key skills

You need to be able to use **directed numbers** in practical situations.

Exam tip

Sketch a number line if necessary and remember to move 25 places to the left.

Questions

1 The average temperature each month in Montreal is shown in the table.

Month	Jan	Feb	Mar	Apr	May	Jun
Avg Temp	−11	−9	−3	5	13	18

Jul	Aug	Sep	Oct	Nov	Dec
21	19	15	8	1	−7

a Find the difference between the highest and lowest average monthly temperatures.

b The average minimum temperature for December is usually 5° lower than the average December temperature. Find the average minimum temperature for December.

2 Calculate the following:

a $17 + (-3)$ **b** $(-4) \times (-2)$ **c** $(-7) - (-1)$

d $15 - (-4)$ **e** $(-18) \div (-3)$ **f** $(-40) \div 10$

3 A man's bank balance is −£345.20. He then withdraws £50. What is his new balance?

4 The water on a gauge is at −20 cm, that is 20 cm below the flood level. If the water rises by 25 cm what level does the gauge now show?

1.5 Number

You need to:

- **Use the language and notation of simple vulgar and decimal fractions and percentages in appropriate contexts.**
- **Recognise equivalence and convert between these forms (including recurring decimals). (Extended)**

If a cake is split into five equal pieces and Joe eats one of the five pieces then the amount he eats can be expressed:

as a **fraction**, that is $\frac{1}{5}$ of the whole cake
as a **decimal**, that is 0.2 of the whole cake
as a **percentage**, that is 20% of the whole cake.

} these three expressions are equivalent

Key skills

You need to be able to simplify fractions.

Exam tip

Divide both the numerator and denominator by the highest common factor of 24 and 30, 6.

Worked example

Write $\frac{24}{30}$ in simplest form. **[2 marks]**

$$\frac{24}{30} = \frac{24 \div 6}{30 \div 6}$$
$$= \frac{4}{5}$$

Key skills

You need to be able to convert between fractions, decimals and percentages.

Exam tip

Multiplying by 100% does not change the value of the number since 100% = 1.

Worked example

(a) Express 0.15 as:
(i) a fraction
(ii) a percentage. **[3 marks]**

(b) Express $\frac{2}{5}$ as a decimal. **[1 mark]**

(a) (i) $0.15 = \frac{15}{100} = \frac{3}{20}$
(ii) $0.15 \times 100\% = 15\%$

(b) $\frac{2}{5} = \frac{4}{10} = 0.4$

Extended

Recap

To convert a recurring decimal to a fraction, multiply the recurring decimal by the power of 10 equal to the number of places of recursion. For example, for 2 place recursion, multiply by 100.

Worked example

Convert $0.\dot{1}\dot{2}$ into a fraction in its simplest form. **[3 marks]**

Let $x = 0.1212...$

$100x = 12.1212...$

$100x - x = 12.1212... - 0.1212...$

$99x = 12$

$x = \frac{12}{99} = \frac{4}{33}$

Key skills

You need to be able to convert from a recurring decimal to a fraction in simplest form.

Exam tip

Multiply x by 100 since it is 2 place recursion.

Subtract x from $100x$ and then simplify the resulting fraction if possible.

Questions

1 Simplify the following fractions.

a $\frac{10}{15}$ **b** $\frac{18}{20}$

c $\frac{35}{40}$ **d** $\frac{120}{150}$

2 Express 0.375 as:

a a fraction

b a percentage.

3 Express $\frac{3}{25}$ as:

a a decimal

b a percentage.

4 Express 55% as:

a a decimal

b a fraction.

5 $0.037 \quad 37\% \quad 0.307 \quad \frac{3}{10} \quad \frac{3}{100} \quad 3.7\%$

From the numbers listed above, write down:

a the smallest number

b the largest number

c the two numbers which are equal.

6 Convert the following recurring decimals to fractions. E

a $0.\dot{2}$

b $0.\dot{6}\dot{1}$

c $0.\dot{1}\dot{3}\dot{5}$

d $0.\dot{6}\dot{1}\dot{6}$